

GANGHUA WANG

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Address: 350 Ford Hall, 224 Church Street SE
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EDUCATION

University of Minnesota, Twin Cities
Ph.D. in Statistics

Aug. 2019 - present
Minneapolis, MN

Peking University
B.S. in Statistics, Minor in Economics

Sept. 2015 - July 2019
Beijing, China

EXPERIENCE

Project: Travelers Modeling Problem
Build prediction models for Peace of Mind Insurance Company

Nov. 2019 - Dec. 2019
Minneapolis, MN

- A competition held in [Kaggle](#). Collaborated with other graduate students, we ranked first in the public leaderboard, and second in the private leaderboard.
- We created several prediction models based on the historical insurance data using Gini coefficient as goodness metric. Boosting model outperforms others like generalized linear model and neural network.

Project: Component analysis in brain image
Supervised by Prof. Hongyu Zhao, Yale University

July. 2018 - Sept. 2018
New Haven, CT

- We proposed to extract functional components from fMRI datasets using Latent Dirichlet Allocation to explore the intrinsic structure of the brain neural network.
- Specifically, we studied the ADHD related neural network and chose a few components. Comparison between our result and other methods like spectral clustering and independent component analysis confirmed that neural edges selected by us are powerful to predict ADHD.

Project: Community detection with co-variates
Supervised by Prof. Jinzhu Jia, Peking University

Dec. 2017 - June 2018
Beijing, China

- We adopted a maximum likelihood estimator(MLE) based method to combine both the information from co-variates and networks to detect communities under the stochastic block model.
- A variational approximation algorithm was developed to solve MLE. Our results beat the state-of-the-art methods like spectral clustering.
- We proved the consistence of both MLE and its variational approximation.

HONORS AND AWARDS

Awarded by University of Minnesota, Twin Cities

Summer Research Fellowship *2020*

School of Statistics First Year Scholarship *2019*

Awarded by Peking University

The Academic Excellence Scholarship (3 times) *2016 - 2018*

Fang Zheng Scholarship *2017*

Wu Si Scholarship *2016, 2018*

Freshman Scholarship *2015*

TEACHING

Teaching Assistant, University of Minnesota, Twin Cities

STAT 4102 Theory of Statistics II *Fall 2020*

STAT 3021 Introduction to Probability and Statistics *Spring 2020*

STAT 3011 Introduction to Statistical Analysis *Fall 2019*

TECHNICAL STRENGTHS

Computer Languages Python, MATLAB, R